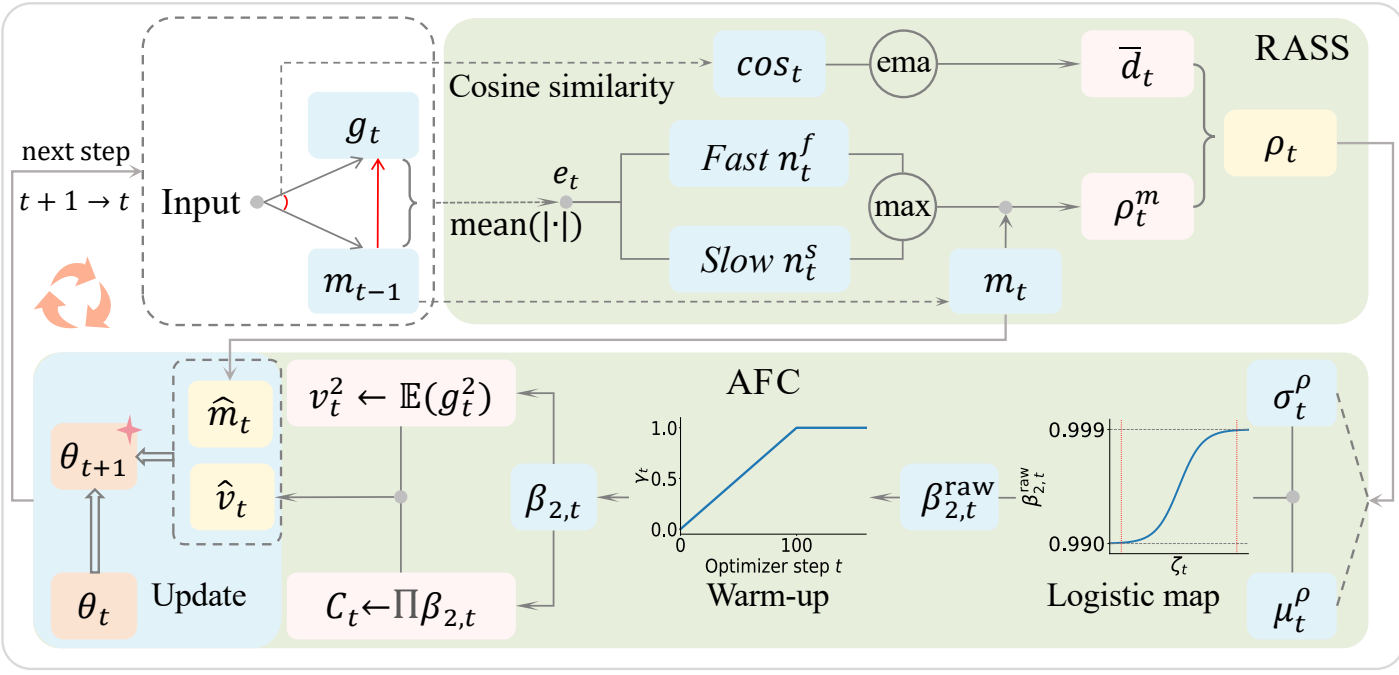
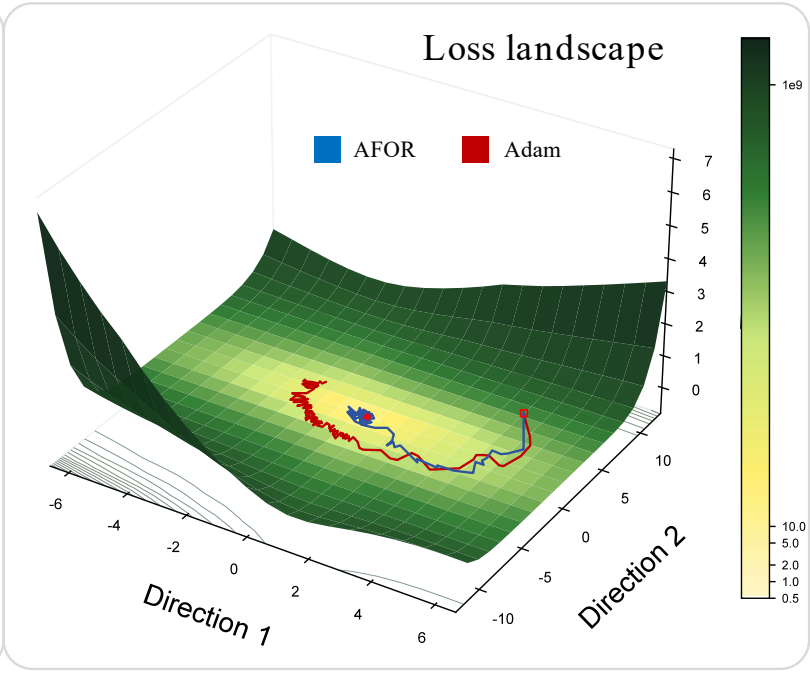

next step
$t+1 \to t$
Input
g_t
m_{t-1}
Cosine similarity
cos_t
ema
$\overline{d}_t$
RASS
ρ_t
e_t
mean(|·|)
$Fast\ n_t^f$
$Slow\ n_t^s$
max
ρ_t^m
m_t
$\hat{m}_t$
$\hat{v}_t$
θ_{t+1}
θ_t
Update
$v_t^2 \leftarrow \mathbb{E}(g_t^2)$
$C_t \leftarrow \Pi\beta_{2,t}$
$\beta_{2,t}$
AFC
1.0
0.5
0.0
γ_t
0
100
Optimizer step t
Warm-up
$\beta_{2,t}^{\text{raw}}$
0.999
0.990
$\beta_{2,t}^{\text{raw}}$
ζ_t
Logistic map
σ_t^ρ
μ_t^ρ


Loss landscape
AFOR
Adam
Direction 1
Direction 2
1e9
10.0
5.0
2.0
1.0
0.5